

Bidual-Complemented Spaces With $\text{se}(X) = 1$ Have the Self-Extension Property

FA Banach automatic math run

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Status. Partial result, likely valid. This proves a bidual-complemented subcase of an open question from Garcia-Gutierrez, Garcia Pacheco, Piniella, and Rambla-Barreno [1].

Source question. The source paper defines a Banach space X to be k -self-extensible if, for every subspace $Y \subseteq X$ and every bounded linear operator $T : Y \rightarrow Y$, one can extend T to an operator $S : X \rightarrow X$ satisfying

$$\|S\| \leq k\|T\|.$$

It then sets

$$\text{se}(X) = \inf\{k \geq 1 : X \text{ is } k\text{-self-extensible}\}.$$

After Corollary 5.4, the paper asks whether X has the self-extension property whenever $\text{se}(X) = 1$.

perplanes (already mentioned after Corollary 4.9).

Finally, let us mention some open questions which appear naturally:

- Suppose the Banach space X has the following property: Whenever Y is a hyperplane of X and $T : Y \rightarrow Y$ is a continuous linear operator, there exists a continuous linear extension $S : X \rightarrow X$ satisfying $\|T\| = \|S\|$. Can we deduce that X has the self-extension property?
- Is there a Banach space X satisfying $\text{se}(X) = \infty$?
- In Corollary 5.3, is it true that $\mathcal{A} = \mathcal{B}$ without any additional hypothesis on X ? In other words, does X have the self-extension property if $\text{se}(X) = 1$?

Idea of the Proof

The finite-dimensional proof in the source passes from almost isometric extensions to an exact extension by compactness of bounded sets in $\mathcal{L}(X, X)$. Infinite-dimensional operator balls are not norm compact, so that argument does not directly survive.

For an arbitrary Banach space X , the canonical embedding $J_X : X \rightarrow X^{**}$ puts the values of almost optimal extensions inside weak-star compact balls of X^{**} . For a fixed subspace Y and operator $T : Y \rightarrow Y$, choose extensions $S_n : X \rightarrow X$ with

$$\|S_n\| \leq (1 + 1/n)\|T\|.$$

The family $(J_X S_n x)_{x \in X}$ has a pointwise weak-star cluster subnet in X^{**} . The cluster point is a norm-preserving extension $R : X \rightarrow X^{**}$. If there is a norm-one projection $P : X^{**} \rightarrow J_X(X)$, then $P \circ R$ lands back in X without increasing the norm. Thus the true obstruction to the arbitrary case is not the compactness step, which always gives an X^{**} -valued operator, but the lack of an exact contractive projection back to X .

Theorem

Theorem 1. *Let X be a Banach space that is 1-complemented in its bidual; namely, suppose there is a norm-one projection $P : X^{**} \rightarrow J_X(X)$, where $J_X : X \rightarrow X^{**}$ is the canonical embedding. If $\text{se}(X) = 1$, then X has the self-extension property. Equivalently, for every subspace $Y \subseteq X$ and every bounded linear operator $T : Y \rightarrow Y$, there is a bounded linear operator $S : X \rightarrow X$ such that $S|_Y = T$ and $\|S\| = \|T\|$.*

Proof. It is enough to consider closed subspaces. Indeed, if Y is not closed and $T : Y \rightarrow Y$ is bounded, then T extends uniquely to a bounded operator $\bar{T} : \bar{Y} \rightarrow \bar{Y}$ with the same norm, and any extension of $\bar{T}$ restricts to an extension of T .

Fix a closed subspace $Y \subseteq X$ and $T \in \mathcal{L}(Y, Y)$. The case $T = 0$ is immediate, so assume $T \neq 0$. Since $\text{se}(X) = 1$, for each positive integer n there is an extension $S_n \in \mathcal{L}(X, X)$ of T with

$$\|S_n\| \leq (1 + 1/n)\|T\|.$$

Put $C = 2\|T\|$. For each $x \in X$, the set

$$K_x = C\|x\|B_{X^{**}}$$

is compact for the weak-star topology $\sigma(X^{**}, X^*)$ by Banach–Alaoglu. Hence the product

$$K = \prod_{x \in X} K_x$$

is compact by Tychonoff's theorem. The points

$$\Phi_n = (J_X S_n x)_{x \in X} \in K$$

therefore have a convergent subnet, say $\Phi_{n_\alpha} \rightarrow \Phi = (u_x)_{x \in X}$ in the product weak-star topology. Along such a subnet, $n_\alpha \rightarrow \infty$ in the usual cofinal sense.

Define $R : X \rightarrow X^{**}$ by $Rx = u_x$. We first check linearity. Let a, b be scalars and $x, z \in X$. For every $x^* \in X^*$,

$$\begin{aligned} u_{ax+bz}(x^*) &= \lim_{\alpha} x^*(S_{n_\alpha}(ax+bz)) \\ &= a \lim_{\alpha} x^*(S_{n_\alpha}x) + b \lim_{\alpha} x^*(S_{n_\alpha}z) \\ &= (aRx + bRz)(x^*). \end{aligned}$$

Since X^* separates points of X^{**} , this gives $R(ax+bz) = aRx + bRz$.

Next, R extends $J_X T$. If $y \in Y$, then $S_n y = Ty$ for every n , hence $Ry = J_X Ty$.

Finally we prove the norm estimate. Fix $x \in X$ and $\varepsilon > 0$. Since $n_\alpha \rightarrow \infty$, eventually

$$\|J_X S_{n_\alpha} x\| \leq (1 + \varepsilon)\|T\| \|x\|.$$

The ball $(1 + \varepsilon)\|T\| \|x\| B_{X^{**}}$ is weak-star closed, so the weak-star limit Rx belongs to it. Therefore

$$\|Rx\| \leq (1 + \varepsilon)\|T\| \|x\|.$$

Letting $\varepsilon \downarrow 0$ yields $\|Rx\| \leq \|T\| \|x\|$ for every $x \in X$, so $\|R\| \leq \|T\|$.

Let $Q = J_X^{-1}P : X^{**} \rightarrow X$, where J_X^{-1} is the inverse isometry from $J_X(X)$ onto X . Define $S = Q \circ R$. Then

$$\|S\| \leq \|Q\| \|R\| \leq \|T\|.$$

If $y \in Y$, then $Ry = J_X Ty$ and $QJ_X Ty = Ty$, so $S|_Y = T$. Hence also $\|S\| \geq \|T\|$, and therefore $\|S\| = \|T\|$. $\square$

Corollary 2. *If X is a dual Banach space and $\text{se}(X) = 1$, then X has the self-extension property.*

Proof. Write $X = E^*$. Then $X^{**} = E^{***}$. The restriction map

$$E^{***} \longrightarrow E^*, \quad \Phi \mapsto \Phi|_E,$$

has norm one and restricts to the identity on the canonical copy of E^* in E^{***} . Thus X is 1-complemented in X^{**} , and the theorem applies. $\square$

Verification Notes and Limitations

The proof uses only Banach–Alaoglu, Tychonoff compactness, weak-star closedness of norm balls in X^{**} , and the assumed norm-one projection from X^{**} onto X . It does not require the extensions S_n to be weak-star continuous; it only takes weak-star limits of their point values after applying the canonical embedding into X^{**} .

This is a partial answer to the source question. It covers all dual Banach spaces, all reflexive spaces, and more generally every Banach space that is contractively complemented in its bidual. It does not settle the same question for arbitrary Banach spaces. It also does not address the two other source questions about hyperplanes or spaces satisfying $\text{se}(X) = \infty$.

Novelty Check

The lightweight run indexes contained no exact entry for arXiv:1911.07826. Local searches for the source title, self-extension property, self-extension coefficient, k -self-extensible spaces, and $\text{se}(X) = 1$ found only the source paper and unrelated occurrences. Web searches for the same terms, with “dual”, “bidual”, and “reflexive” variants, found the source paper [1] and the earlier Lyapunov/self-extension paper [2], but no exact later answer to the bidual-complemented subcase. Novelty confidence is therefore bounded rather than absolute.

Human Review Recommendation

Check that the definition of subspace intended in the source permits the initial closure reduction. If the source convention silently restricts to closed subspaces, the reduction is unnecessary and the proof begins after it. The main compactness step should be reviewed by verifying that the chosen subnet has indices tending to infinity, so that the final norm of the X^{**} -valued extension is exactly $\|T\|$, not merely $2\|T\|$. The final projection step is exactly where the arbitrary Banach-space case remains open.

References

- [1] J. B. Garcia-Gutierrez, F. J. Garcia Pacheco, P. Piniella, and F. Rambla-Barreno, *Non-Hilbert Banach spaces with the self-extension property*, arXiv:1911.07826, 2019/2021.
- [2] K. Loskot, A. Polanski, and R. Rudnicki, *Further comments on: Vector norms as Lyapunov functions for linear systems*, IEEE Transactions on Automatic Control 43 (1998), no. 2, 289–291.
- [3] W. Rudin, *Functional Analysis*, 2nd ed., McGraw–Hill, 1991.